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Syngas Membrane Engineering Design and Scale-Up Issues. Application of Ceramic Oxygen Conducting Membranes

Michael Carolan

The use of ceramic membranes to produce synthesis gas (or syngas), a mixture of hydrogen and carbon monoxide, is a very exciting and novel application for a ceramic material. For successful commercialization, the ceramic body used for the syngas membrane is required to achieve a high oxygen flux, have a very low leak rate, preferably withstand a high pressure differential from the syngas side of the membrane to the air side of the membrane, achieve a high reliability, operate at high temperature, and attain a long lifetime while also being economic to manufacture. This is breaking new ground for a ceramic material. There are no parallel ceramic technologies in commercial practice today with all of these requirements, at best, there are a few commercial ceramic technologies with some, but not all, of these requirements. For example, ceramic filtration membranes are used under a high pressure differential, but they are not made from ion conducting ceramics, nor do they generally operate at high temperatures. Another example is a solid oxide fuel cell which is made from an ionic conductor, but which is not required to operate under a high pressure differential. As they design, fabricate, and test their membranes, the developers of ceramic membranes for syngas production are blazing new trails in ceramic engineering and science. As a consequence, a number of innovative technologies have been developed, which are accelerating the development of ceramic syngas membranes and which may even have application in areas beyond ceramic membranes.

Besides the development of the ceramic components of the membrane reactor, the commercialization of ceramic membrane technology requires the design of specialized reactors to house the ceramic membranes, formulation of ceramic-to-ceramic sealing technology to join the ceramic components of the membrane, and invention of ceramic-to-metal seals to attach the membranes to metal manifold pipes, all tailored to the unique and demanding requirements of the syngas production process. Many different technical disciplines, including ceramic engineering, chemical engineering, and mechanical engineering are required to address all of these challenges. This was one of the drivers that led to the formation of a number of large industrial and government partnerships to develop ceramic membranes. One consortium, led by Air Products and Chemicals, Inc. and including the United States Department of Energy, Ceramatec, Inc., Chevron

Nonporous Inorganic Membranes.
Edited by Anthony F. Sammells and Michael V. Mundschau
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ISBN: 3-527-31342-7

Energy Technology Co., Eltron Research Inc., SOFCo EFS, The Pennsylvania State University, Norsk Hydro, Pacific Northwest National Laboratory, and the University of Pennsylvania, is developing the ion transport membrane (ITM) syngas membrane process. A second consortium, led by Praxair, Inc. and including the United States Department of Energy, BP, Foster Wheeler, Sasol and Statoil, is developing the oxygen transport membrane (OTM) syngas membrane process. Teikoku Oil, Nippon Steel, NKK and METI have formed a similar consortium in Japan to develop ceramic membranes for syngas production. A similar feature of all three consortia is that they are formed from a variety of organizations with complementary technical skills. The multidisciplinary challenges that need to be overcome to commercialize ceramic syngas membranes can only be faced with a technical team whose skills are well matched to the challenges. This chapter focuses on some of the aspects of this technology that are required to scale ceramic membranes from the laboratory to commercially relevant sizes: ceramic membrane design and engineering, reactor design, ceramic-to-ceramic seals and ceramic-to-metal seals.

8.1 Membrane Design and Engineering

Processes using ceramic membrane technology must achieve a significant cost benefit in order to displace the conventional technologies that currently exist to produce hydrogen and syngas. A number of factors will contribute to the economics of a ceramic membrane process, including the performance of the membrane, the cost of the membrane, the cost of the reactor containing the membrane and the cost of the balance of the plant. Achieving a high oxygen flux is central to the performance of a mixed conducting ceramic membrane. This is driven primarily by economic considerations: the higher the oxygen flux, the less membrane area that is required, which in turn reduces the capital cost of the ceramic membrane process. Reducing the required membrane area also reduces the size of reactor vessel that holds the ceramic membranes, again reducing capital cost. Of all of the aspects of membrane performance, the oxygen flux has obtained by far the most attention in the literature. Oxygen flux values greater than 10 sccm O_2 cm^{-2} are routinely quoted in the literature [1]. Various means have been employed to increase the oxygen flux. Increasing the operating temperature increases the oxygen flux because the oxygen anion transport through the membrane is thermally activated. However, there are limitations to this approach because the mechanical properties of the membrane, in particular the creep rate, degrade with increasing temperature [2, 3]. The creep rate will generally have a much higher activation energy than the oxygen anion diffusivity and therefore will generally place an upper limit on the maximum operating temperature of a ceramic membrane. Another approach to increasing the oxygen flux is to decrease the membrane thickness. Some authors have reported oxygen flux as increasing with the inverse of the membrane thickness [4]. Other authors have found that the flux is not inversely

proportional to the membrane thickness and attributed this to slow interfacial reaction kinetics or to gas phase mass transfer resistance on one or both sides of the membrane [5]. The limiting resistance is a function of the membrane composition, thickness and operating conditions. By adding a porous layer on one or both sides of the membrane to increase the surface area of the membrane, the surface kinetic limitation can be reduced, allowing higher fluxes to be achieved [6]. Even if the surface kinetic and gas phase mass transfer limitations can be overcome, decreasing the membrane thickness creates significant challenges of how to support a large pressure differential across a thin membrane and how to avoid leaks through defects in the thinner membranes. Developers of ceramic syngas membranes must take these factors into consideration when designing their membranes.

One of the significant challenges presented by the use of ceramic membranes to produce syngas is the large pressure differential across the membrane. The permeate side of the membrane is exposed to the reducing methane-containing stream while the other side of the membrane is exposed to the oxidizing conditions of the air feed. The methane-containing stream will usually be available at high pressure, >200 psig (1.4 MPa). The process economics are also most favorable if the methane stream is kept at high pressure, to integrate well with any downstream gas-to-liquids (GTL) or hydrogen separation processes, which will require a pressurized syngas feed. Conversely, it is desirable to use a low pressure air feed to reduce compression costs. This results in a large pressure difference across the membrane and requires membrane material properties and a membrane design capable of withstanding such a pressure difference, both in terms of strength and in creep resistance. The mechanical design must limit the maximum stresses, in particular tensile stresses, that the membrane experiences during steady state and transient operation. Decreasing the stress decreases the probability that the membrane will fail and hence increases the mechanical reliability of the membrane. Decreasing the stress will also decrease the creep rate. Creep is a potentially lifetime limiting phenomena. The creep rate will be a power law function of the stress, with the power law stress exponent a function of the specific membrane composition used [3]. Therefore, lowering the stress will lower the creep rate and potentially extend the membrane life. Designing the membrane becomes an exercise in balancing the competing demands of increasing the flux and decreasing the tensile stress in the membrane. Finding an acceptable membrane design that can also meet the economic goals for the ceramic membrane process is the challenge which all of the developers of this technology must meet.

Some of the unique properties of mixed ionic and electronic conducting ceramics can present significant membrane design challenges that are not found with conventional ceramic materials. The large oxygen chemical potential gradient across the dense membrane during operation can produce significant differential strain between the air and methane sides of the membrane due to chemical expansion of the membrane, as described by Adler [7] and Hendriksen et al. [8]. The low oxygen partial pressure, syngas side of the membrane expands relative to the air side of the membrane due to the lower oxygen stoichiometry on the low

oxygen partial pressure side of the membrane. This will produce tensile stresses on the air side of the membrane and compressive stresses on the syngas side of the membrane. Depending on the membrane composition and membrane design, the magnitude of the stresses produced by the differential expansion can be significant. Solutions have been proposed to address this problem. One solution suggests adding second phases to the membrane to improve the mechanical properties of the membrane [9, 10]. The second phases can either improve the strength of the membrane or, if the second phases are ductile, the ductile phase can accommodate the chemical expansion strain in the ceramic phase by deforming. Another solution that has been proposed is to produce a two-phase material where the expansion of the first phase is counterbalanced by the contraction of a second phase [11]. A third solution that has appeared in the patent literature tailors the membrane composition to produce an acceptably low chemical expansion [12, 13]. No matter what solution is adopted, the stresses produced by chemical expansion must be considered when designing the membrane structure to maintain acceptably low stresses during steady state and transient operation and hence a high mechanical reliability.

Two basic approaches to membrane design have been pursued by practitioners in this field: the tubular design and the planar design. Each design has its own set of advantages and its own set of challenges. Much of the very early basic research in this field was performed using small planar disk membranes. These are easy to manufacture on the laboratory scale. Notable examples are the work of Teraoka et al. [14] in Japan and Mazanec and coworkers at Standard Oil of Ohio [15] who performed the ground breaking work in this field. These planar membrane designs were suitable for demonstrating the potential of the technology, including demonstrating high oxygen fluxes. However, these designs were generally not capable of supporting a significant pressure differential nor were these simple disks amenable to scaling to larger sizes. Different designs were needed to advance the technology.

One of the first routes taken to overcome the challenges of achieving a pressure differential between the air and methane sides of the membranes along with obtaining more membrane area, was the tubular design. Self-supporting tubes can be fabricated relatively easily without sophisticated equipment, either by slip casting or extrusion. These methods yield thick-walled tubes, typically with a wall thickness of 0.5 mm or greater. Using such a design, tubular membranes have been tested for over 6 months at a pressure differential of 250 psig (1.7 MPa) at 850 °C [16]. Fabrication of tubular membranes of ionic conductors up to 3 feet (0.9 m) in length have also been demonstrated, showing the facility of scaling up membrane size using a tubular membrane design. Tubular membranes have shown the capability to operate at high pressures and to be scaled to larger sizes. However, to achieve all of the potential benefits of ceramic membrane technology, further improvements are needed.

Although self-supported tubular designs have certain advantages in terms of ease of fabrication, it is difficult to meet simultaneously the requirements of a thin separating layer to achieve a high oxygen flux and a sufficiently thick wall to

withstand a pressure difference. Developers have attempted to meet this challenge by using supported thin film membranes. Examples of such designs include thin dense membranes on porous supports. One of the earliest examples of such a membrane was the work of Hazbun at Arco Chemical [17]. This membrane consisted of a 50 μm dense membrane of titania-doped yttria-stabilized zirconia on a porous magnesia-stabilized zirconia support. The thin separating layer was deposited by electrochemical vapor deposition. One common feature of these membranes was that the porous support and the membrane were made from similar compositions. Using similar compositions for the membrane and the support ensures good thermal expansion match and chemical compatibility of those components.

Although using the same material for both the separating layer and the support layer has obvious advantages, there are also disadvantages to that design philosophy which have driven some developers to produce tubular supported thin film membranes with dissimilar materials for the support layer and the separating layer. Using dissimilar materials opens up the possibility of independently choosing the support material and the dense separating material with the properties best suited for each of those roles. For example, the membrane material is chosen to have high oxygen flux rates while the support material is chosen for its superior mechanical properties. A U.S. patent describes such a membrane developed by BP Amoco [18]. In this invention, porous tubes of 316 stainless steel, Incoloy 800HT or Haynes 230 nickel superalloy, were used as porous supports for thin dense ceramic membranes of $SrCoFe_{0.5}O_x$. The ceramic membrane was applied to the metal porous supports by thermal spraying of a fine powder onto the supports. The resulting membranes were able to withstand pressure differentials of up to 100 psig (0.7 MPa) using inert gas and up to 50 psig (0.34 MPa) under syngas production conditions. For application in syngas service, improvements would be needed to allow higher pressure differentials to be used.

One challenge faced by tubular membranes is creep buckling of the tubes when under external pressure. Any imperfection in the circularity of the tubular membrane crosssection or wall thickness results in a nonuniform stress profile in the tube. Creep will cause the membrane to further deform and accentuate the noncircularity, further increasing the maximum stress and accelerating the creep. In a self-reinforcing and self-accelerating process, a tube could eventually buckle if the creep rates are high enough. A method has been suggested in the patent literature to address this phenomenon [19], as shown in Fig. 8.1, porous structural members (labeled 12 and 14 in the figure) are placed inside the membrane (labeled 10 in the figure) to limit the amount of deflection the membrane can sustain. While preventing creep buckling, other means would be necessary to accommodate the air flow inside the membrane.

The second category of membrane designs which has enjoyed wide scrutiny is flat plate membranes. Under the heading of flat plate membranes, two varieties have been investigated. Self-supporting, flat membranes have been used by a wide number of researchers in small scale experiments. However, this type of membrane has limited potential for scaling up and for application under high pressure

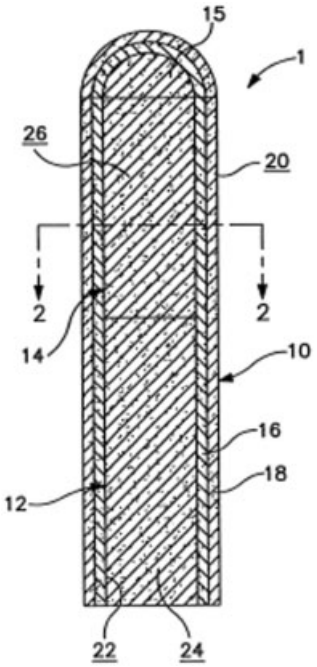


Figure 8.1 Figure 1 from U.S. Patent 6 565 632 B1 [19] showing porous support members 12 and 14 to prevent the creep buckling of the tubular membrane 10 under an applied external pressure load.

differentials. The second type of membrane design uses microchanneled flat plate membranes that have been developed by a small number of industrial consortia as they have scaled-up membrane and reactor dimensions. Microchannel flat planar membranes have the capability to address the most significant challenges presented by the ceramic syngas membrane process.

Microchanneled designs have significant benefits which have the potential to more than pay back the investment in their more complex design relative to tubular membranes. Figure 8.2 is a schematic that illustrates some of the key features of the microchanneled design. Air is delivered to a dense separating membrane through multiple parallel channels. Syngas flows in the gaps between adjacent membranes. Some of the potential advantages of the microchanneled designs include: very low gas phase mass and heat transfer resistance on both the air and syngas side of the membrane, high ratios of membrane surface area per unit volume of the reactor, and a decreased number of ceramic-to-metal seals per unit of membrane area. Each of the advantages can lead to a potential performance and cost advantage for planar membranes over tubular membranes. Taken together, these advantages can lead to a very significant cost advantage for planar membranes.

Gas phase heat and mass transfer resistance becomes increasingly important as higher flux membranes, such as supported thin film membranes, are developed. At high fluxes, the rate at which oxygen from the feed air can diffuse to the membrane surface or the rate at which the syngas reactants can diffuse to the mem-

brane surface can be the rate limiting step for oxygen transport across a syngas membrane if care is not taken to minimize the gas phase mass transfer resistances on both sides of a membrane. Equally important to good mass transfer is efficient heat transfer from the membrane to the gas phase. The oxidation reactions which occur on the membrane surface are exothermic. If this heat cannot be transferred efficiently to the air and syngas gas streams, the membrane will become hotter than the gas streams on either side of the membrane, potentially becoming heated to temperatures where degradation phenomena such as creep can occur. The low gas phase heat and mass transfer resistances in the microchanneled design are due to the small dimensions of the gas channels where the air and syngas flow. As the dimensions of the gas channels adjacent to the membrane surface become smaller, the distance that gas molecules must diffuse from the bulk gas flow to the membrane surface grows smaller. The smaller diffusion distances increase the rate at which reactants can reach the surface and hot gas molecules can diffuse away from the surface. On the high pressure, syngas side, the membranes can be spaced sufficiently close to minimize the gas phase mass transfer resistance for the reactants in the syngas to reach the membrane surface. On the low pressure, air side of the membrane, the height of the channels inside the membrane can be sized to minimize the diffusional resistance of oxygen reaching the membrane surface. There are limits, of course, on how close the membranes can be spaced and how small the air channels can be, as pressure drop becomes an important consideration as channel dimensions decrease. Deter-

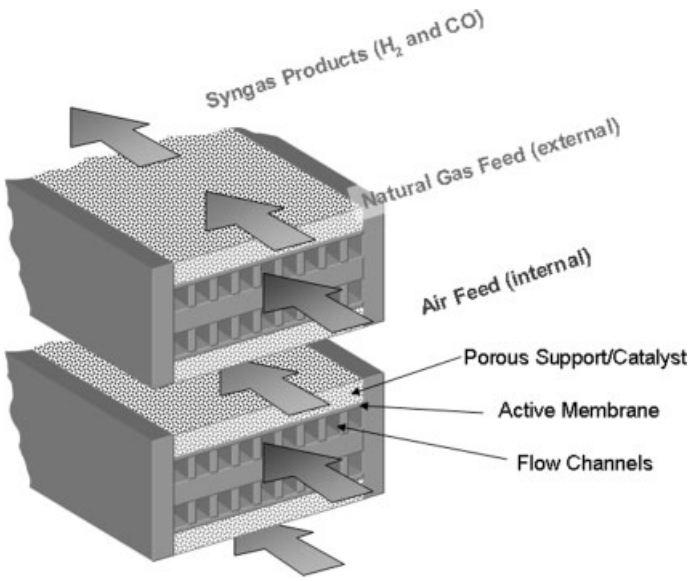


Figure 8.2 Planar microchanneled ceramic membrane. Syngas flows over the exterior of each membrane while the air flows in internal channels inside each membrane.

mining the wafer spacing and channel dimensions becomes an economic optimization balancing the competing factors of increasing gas compression costs versus higher fluxes and less membrane area.

It is difficult to obtain the same high gas phase heat and mass transfer with tubular designs as with microchanneled planar designs. On the internal, air side of a tubular membrane, one would need to either decrease the inside diameter of the tube to the dimensions of a microchannel or introduce an internal structure to the tube to produce microchannels. Either approach results in a membrane design that could have a high air side pressure drop per linear length of membrane. This would require the use of shorter tubes to produce reasonable air side pressure drops. This comes at the cost of increasing the number of tubes and hence increasing the number of ceramic-to-metal seals and air side manifolding, which would increase the capital cost of such a design. Flat plate microchanneled designs, which have a large number of microchannels in parallel, but for which the length of the microchannels is only the length of each membrane, would not suffer from this limitation.

For an array of tubular membranes, the gas phase mass and heat transfer on the external, syngas side of the membrane is inversely proportional to the distance between nearest neighbor tubes. The requirements of forming a seal to a tube sheet at one or both ends of each tube make it difficult to decrease this distance to much less than the tube diameter or even a multiple of the tube diameter, based on the experience of shell and tube heat exchangers. Again, decreasing the tube diameter would improve the gas phase heat and mass transfer but at the cost of increasing the number of tubes to achieve the same membrane area with the concomitant increase in the number of ceramic-to-metal seals. For planar membrane designs, the syngas side membrane spacing can be fabricated to the desired dimension to obtain the required heat and mass transfer with only the pressure drop in the syngas streams as a consideration.

Besides the advantages of efficient syngas side heat and mass transfer, the close packing of the membranes afforded by planar designs produces the benefit that the amount of membrane area per unit volume of reactor can be significantly larger for planar designs than for tubular designs. This results in a decreased size of the reactor vessel and hence a lower cost vessel relative to a tubular design of the same membrane area. A parallel benefit is the decreased number of ceramic-to-metal seals needed for a planar membrane design than for a tubular design of the same membrane area. Each tube in a tubular design will require one high pressure ceramic-to-metal seal for a bayonet design and two seals for a flow through design. In a planar design, individual membranes can be stacked into a multimembrane module, therefore, only one high pressure ceramic-to-metal seal would be required at the base of a stack of membranes with a considerable membrane area. If high pressure ceramic-to-metal seals are a significant cost component of the syngas reactor, then using a reactor with planar membranes has the potential to lower the cost relative to a reactor containing tubular membranes.

An example of the internal structure of the microchanneled, flat plate membranes developed by Air Products and its partners Ceramatec and SOFCo EFS, is

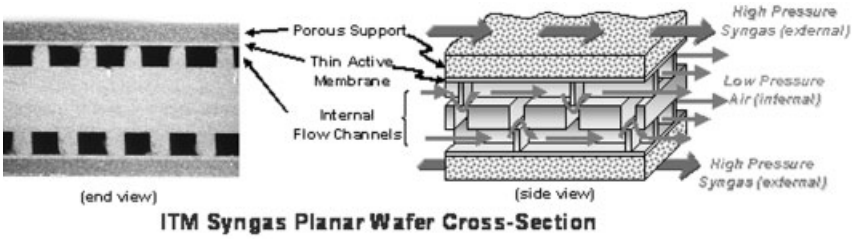


Figure 8.3 Internal structure of the ITM Syngas Membrane developed by Air Products and its partners Ceramatec and SOFCo EFS.

shown in Fig. 8.3 [20]. The membrane is a multilayer laminate that is symmetric about the midplane of the membrane wafer. Porous layers form the top and bottom surfaces of the membrane. These porous layers are exposed to the high pressure syngas, and they are adjacent to the dense separating layer. The porous layers help support the dense layer. The internal structure of the membrane consists of multiple channeled layers to direct the air flow. The channeled layers also provide mechanical support for the dense separating layers against the pressure load. By keeping the high pressure gas on the outside of the membrane, the membrane is kept in hydrostatic compression. The compressive strength of a ceramic typically exceeds the tensile strength by a factor of five. Therefore, designing a ceramic membrane to keep the membrane components in compression rather than tension is an important goal. The channels in the two channeled layers next to each of the dense layers provide air feed to the separating layers. Another channeled layer is sandwiched between these two channeled layers. This middle channeled layer has channels perpendicular to the channels of the first two layers which allow redistribution of the air flow between the channels of the first two layers. By careful design of the flow resistance in the channel layers, even flow distribution of the air over the entire membrane can be achieved.

To achieve all of the benefits of the microchanneled, flat plate design, multiple membrane wafers must be stacked together to form multiwafer modules.

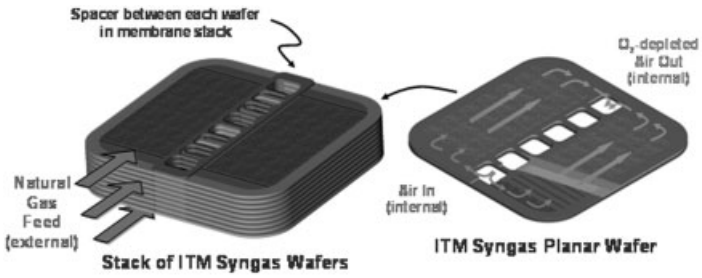


Figure 8.4 A membrane module consisting of a stack of planar microchanneled membranes (left) and the air flow path in each membrane (right).

Figure 8.4 shows a conceptual picture of a stack of the membrane wafers and the overall flow patterns for the air and syngas streams [20]. Spacers are placed between each membrane wafer in the stack. The spacers define the gap between each membrane and set the height of the syngas flow channel. Air and nonpermeate manifold holes pass through each spacer and wafer. These manifolds serve to supply air feed to each membrane in the stack and also to collect the nonpermeate from each membrane. Flow distribution channels at the leading and trailing edges of each membrane wafer serve to evenly distribute the air flow across the width of the membrane while keeping the overall pressure drop across the membrane to a minimum.

Of the various planar membrane designs that have appeared in the patent literature, development of the design from the Air Products consortium appears to have advanced the most. They have scaled up membranes from the laboratory scale to full commercial size as shown in Fig. 8.5 [21]. The membrane area has been scaled up by a factor of over 300 from the laboratory scale to the commercial scale membrane. Pilot scale membranes have been operated in a process development unit at pressures up to 425 psig (3 MPa) and temperatures up to 900 °C at commercial fluxes. Although other organizations have reported designs of planar membranes, there have been no reports from these organizations on the fabrication of planar membranes nor on the operation of planar membranes under commercially relevant conditions.

Monolithic membrane designs are appealing since, potentially, it is possible to produce a body with a significant membrane area, much like a planar membrane stack, but to reduce the need for the ceramic-to-ceramic seals needed to construct the planar membrane stack. Several concepts for monolithic ceramic membrane reactors have appeared in the patent literature. The earliest concept was developed by Hazbun of Arco Chemical and published in 1988 [17]. Figure 8.6 shows two membrane design concepts from the patent including a corrugated membrane

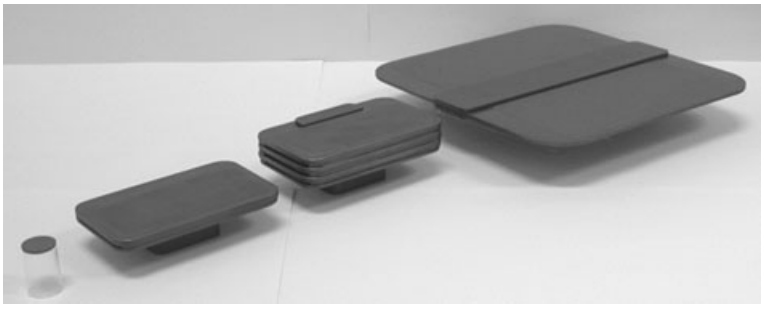


Figure 8.5 The progression of scale up of the Air Products syngas membranes, from laboratory sizes membrane at the left, through a single pilot scale membrane, a three wafer high stack of pilot scale membranes to the commercial size membrane at the right.

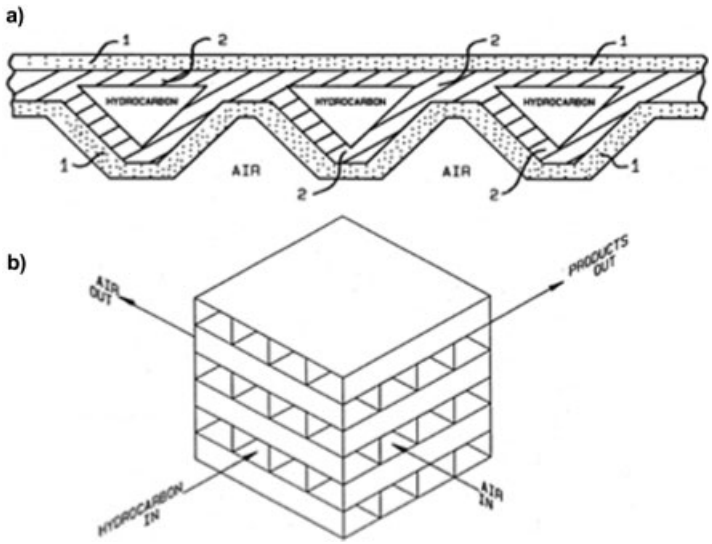


Figure 8.6 Figures 6 (A) and 7 (B) from Hazbun, U.S. Patent 4 791 079 [17] showing some of the first concepts for a monolithic ceramic membrane design for the partial oxidation of hydrocarbons.

structure and a cross flow reactor. Although Hazbun was the first to publish these concepts in a patent, there is no evidence that either of these concepts was fully developed or implemented.

Another cross flow monolithic design for a ceramic membrane was described in a patent assigned to Amoco [22] and is shown in Fig. 8.7. The design was adapted for either a series of membrane cores or a monolithic membrane. The air and reactant feeds are in cross flow with manifolds provided for both the air and reactant gas streams. The monolithic design shown on the right of Fig. 8.7 has the potential to minimize the number of seals, because the only seals required are between the monolith and the manifolds. However, this reduction in the number of seals comes at the expense of greatly increased ceramic fabrication complexity, since producing a monolithic body with cross flow channels is a ceramic fabrication challenge. While the manifolds of the monolithic membrane from this patent [22] are simple in design, the monolithic ceramic membrane body is complex. Fabrication of a large, thick ceramic body with perpendicular internal features is likely to be difficult.

A different approach to monolithic membrane design has been taken by Norsk Hydro [23]. They chose to make a less complex monolithic membrane body while the manifolds applied at the ends of the monolith are more complex than the Amoco design of Fig. 8.7. The Norsk Hydro monolithic membrane design is shown in Fig. 8.8. In this design, the monolithic body consists of parallel straight channels for the oxidant and reactant gas flows. The oxidant and reactant channels are adjacent to each other, in alternating rows or in a checkerboard pattern in

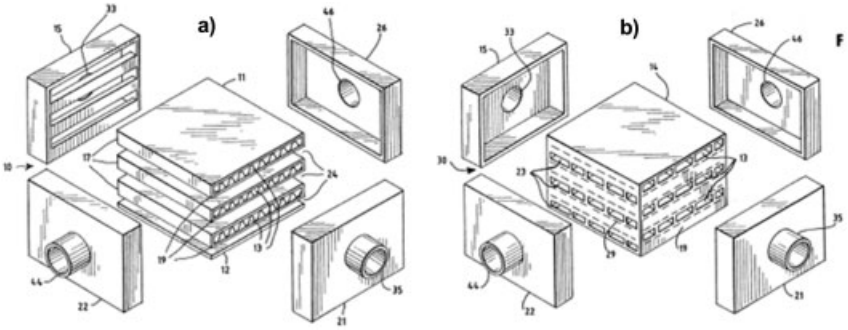


Figure 8.7 Figures 1 (A) and 3 (B) from U.S. Patent 5 356 728 [22] showing a cross flow monolithic reactor (10 and 30) from Amoco consisting of membrane cores (A, 17) or a monolithic block (B, 19). The manifold (15) distributes either the oxidant or reactant to the cores, while the other feed manifold (22) provides the other gas. Manifolds (21) and (26) are the exhaust manifolds.

the monolith. This design could simplify the manufacture since, for example, very similar ceramic structures are currently commercially produced for catalyst carriers for automotive catalytic converters. The oxidant and reactant flows in the Norsk Hydro design can either be co-current or countercurrent. Only two manifolds are required, one at each end of the monolith. However, the manifolds in this design must distribute or collect both the oxidant and reactant gases. For example, a manifold at the first end of the monolith would feed the oxidant gas

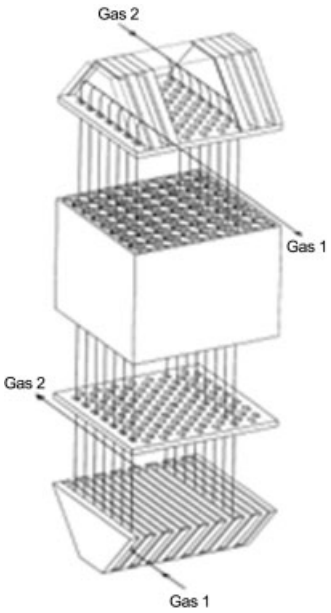


Figure 8.8 The Norsk Hydro monolithic membrane design from Fig. 7 of WO03/033985 [23] showing the monolithic membrane body with top and bottom manifolds for feeding and withdrawing two gases.

into a first set of monolith channels and collect the syngas product from a second set of monolith channels. This design will require a robust sealing technology since there is potentially a large amount of seal area between each manifold and the monolith. Manufacturing the manifolds themselves may also be challenging due to the complex three-dimensional features required.

8.2 Reactor Design and Engineering

Developers of ceramic syngas membranes also face unique challenges on how to deploy their membranes in a practical and economic reactor as they scale up their processes. These challenges arise for many of the same reasons that posed the unique challenges for the membrane design: the high operating temperature of the membranes; the high pressure differential between the high pressure syngas and the low pressure air; and the need to achieve a compact design for economic reasons. The reactor must also have separate inlets and outlets for the syngas and air streams. The reactor designs that have appeared in the patent literature and elsewhere have been driven by the developer's choice of membrane designs. The reactors for tubular membranes and the reactors designed for planar membranes have evolved to accommodate the unique features and design constraints of each type of membrane. The resulting reactor designs are very different from any equipment currently in use in the chemical process industries. Reactor design is another area where very innovative thinking has been applied to tackle the interdisciplinary challenges presented by ceramic membrane technology.

Several designs for reactors for tubular membranes have appeared in the patent literature. Reactors designed for tubular membranes share a number of features with shell and tube heat exchangers. This design philosophy makes sense when one considers the syngas reactors as a device for mass transfer, much as a heat exchanger is a device for heat transfer. Shell and tube heat exchangers are a well developed, mature technology and translating the proven design features of the heat exchanger to the ceramic membrane reactor could reduce the technical risk of the ceramic membrane reactor. Reactor designs for tubular membranes using either a bayonet design or a flow through design have been suggested in the patent literature. These designs replace the metal tubes of a heat exchanger with the ceramic membrane tubes to make a syngas reactor. Additional features have also been added to the syngas reactor designs to address some of the unique challenges that arise when applying ceramic membranes for syngas production, including the need for ceramic-to-metal seals, the desirability to transfer heat in addition to mass between the reactant streams and the possibility to include heterogenous catalyst beds in the reactor.

A prototypic design for a tubular membrane reactor can be found in Linde AG's patent application shown in Fig. 8.9 [24]. This design uses a bundle of ceramic membrane tubes that are inside a pressure vessel. The tubes are closed at one end with the air fed to the tubes in the annulus between the ceramic tube, labeled 32,

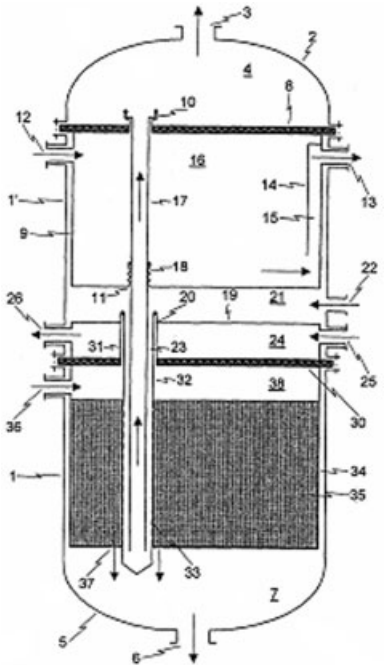


Figure 8.9 Figure 1 from DE 10 114 173 A1 [24], showing the ceramic membrane tube 33. Fresh syngas is supplied at port 36 and syngas product is withdrawn at port 6.

and a bayonet-style air withdrawal tube, labeled 17. Fresh air enters at port 22 and spent air is withdrawn at port 3. Fresh reactant gas is supplied at port 36 and withdrawn at port 6. Two separate tube sheets are required, the first for the ceramic syngas membrane tube, labeled 30, and the second for the air withdrawal tubes, labeled 8. If the air is supplied at a lower pressure than the syngas feed, then the tube sheet holding the ceramic membranes would be subjected to the full pressure differential between the air and syngas streams. Provision is also made for conducting heat exchange between the hot process streams inside the vessel. Port 25 is provided to introduce either cold air or steam into space 24. The air or steam is heat exchanged against the hot process gas to preheat the air to be used as the feed for the process or to raise the steam temperature so it can be mixed with the hydrocarbon feed to the reactor. The preheated gas is withdrawn through port 26. If steam is preheated in space 24, then space 16 in the reactor can be used to preheat the air feed against the hot oxygen depleted air stream in the bayonet tubes. As stated in the patent, this design minimizes the amount of ceramic components that must be used in the reactor and uses as much conventional technology as possible. This can reduce the technical risk during scale-up and development. However, as will be explained below, this type of design may have limitations that could be disadvantageous as the technology is scaled to larger sizes.

As noted in another patent, also from Linde AG [25], one of the disadvantages to their previous design [24] is the unfavorable aspect ratio of the pressure vessel for this reactor design. To scale up ceramic membrane technology and to increase

the membrane area in a reactor, the ceramic membrane tubes can either be made longer or more tubes can be added to a reactor. However, the length of the ceramic membrane tubes will be limited by practical ceramic fabrication considerations to a reasonable length, for example, 2 m. It would be difficult to make significantly longer tubes with the requisite straightness. Therefore, the only way to scale up the reactor size is to increase the number of membrane tubes, which would then require increasing the diameter of the pressure vessel. However, as the diameter of the pressure vessel increases, the required thickness of the vessel wall will increase with the vessel diameter to maintain the same maximum allowable working pressure. Therefore, the mass, and hence the cost of the pressure vessel, increases rapidly as the reactor diameter increases. Another limitation of this design is the requirement for internal tube sheets. Given that at least the tube sheet separating the air and syngas atmospheres has a high pressure difference across it, and given that the tube sheet is exposed to hot gas on both surfaces, this tube sheet would need to be made from an expensive high temperature alloy and be of a very heavy gauge to withstand the pressure difference, further adding to the mass and expense of the vessel. The required thickness of the tube sheet will also increase with increasing vessel diameter. Due to these factors, reactor designs using tubular membranes with the tubes oriented parallel to the axis of the vessel suffer from less than ideal scaling factors. To increase the membrane area by adding more membrane tubes, the reactor vessel becomes proportionately heavier and hence more expensive. At some point, it would be more economic to consider multiple parallel reactors instead of a single large reactor.

To address the concerns raised above, Linde AG has proposed an alternate reactor design with the axis of the membrane tubes oriented perpendicular to the long axis of the reactor vessel [25]. This alternate design is shown in Fig. 8.10. Tubes or bundles of membrane tubes are inserted into the vessel through the penetrations, labeled 9 on the drawing, on the diameter of the vessel. Additional membrane area is simply added by making the vessel longer and adding additional penetrations for additional membrane tube bundles. There are also no large diameter, internal tube sheets in this design, avoiding the potential expense of that component. This type of reactor design, with tubes oriented perpendicular to the axis of the vessel is thus easily scaled to larger capacities while maintaining favorable reactor aspect ratio. Also, the higher aspect ratio of this design is more favorable for shipping the vessel, since there are often limitations on the diameter of a vessel that can be shipped by road or rail. A third stated advantage of this design is that, because the syngas flow is perpendicular to the tube axes, the mass and heat transfer from the gas phase to and from the membrane surface will be enhanced over the design in which the syngas flow is parallel to the membrane tube axes. Although the flow geometry is complex, and many factors come into play, heat and mass transfer coefficients for flow perpendicular to a tube or bundle of tubes are generally higher than the heat and mass transfer coefficients for flow parallel to a tube, mainly due to the mixing produced as the gas flows around the tube. A fourth advantage of the reactor design of Fig. 8.10 is the increased ease in replacing a portion of the membrane surface area relative to the reactor shown in

Fig. 8.9. A bundle of tubes can be easily accessed and replaced through the penetrations on the diameter of the reactor shown in Fig. 8.10. Replacing tubes or bundles of tubes for the reactor shown in Fig. 8.9 is unlikely to be as straightforward. Linde AG also claims improved controllability of the reactor since oxygen is added to the syngas stream incrementally by each tube bundle. Conversely, in the design shown in Fig. 8.9, the syngas would be fully converted as it traveled along the length of each tube. This could produce a significant temperature gradient along each tube since the net methane partial oxidation reaction is exothermic. Therefore, the reactor design with the membrane tubes oriented perpendicular to the gas flow offers a number of advantages and differences from the design with the tubes oriented parallel to the gas flow.

One possible disadvantage of the reactor design shown in Fig. 8.10 is that it requires a considerable number of vessel penetrations. The number of penetrations will be proportional to the membrane area. Since each penetration will need to be cut into the pressure vessel and then a chamber welded onto the vessel at each cut, the use of multiple penetrations could add significantly to the cost of the vessel.

Praxair has also filed patent applications on a syngas reactor incorporating tubular membranes, as shown in Fig. 8.11 [26]. Similar to the Linde AG design of Fig. 8.9, this design uses tubular membranes oriented with the axes of the tubes parallel to the axis of the cylindrical pressure vessel. High pressure syngas flows on the exterior of the ceramic membrane tubes, and low pressure air is fed into

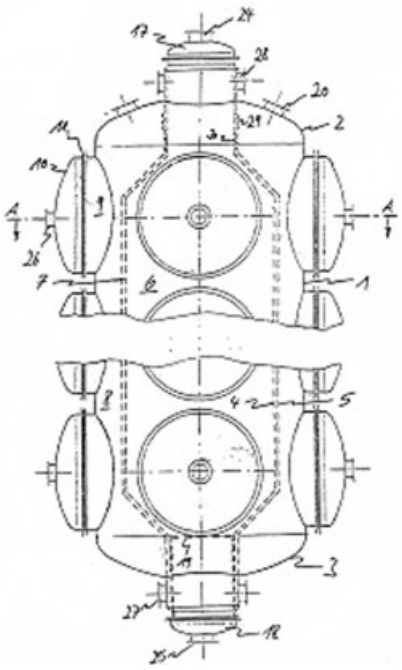


Figure 8.10 Figure 1 from DE 10 056 787 A1 showing a reactor for tubular membranes where bundles of tubes are inserted into the reactor through penetrations (9) on the diameter of the pressure vessel (3).

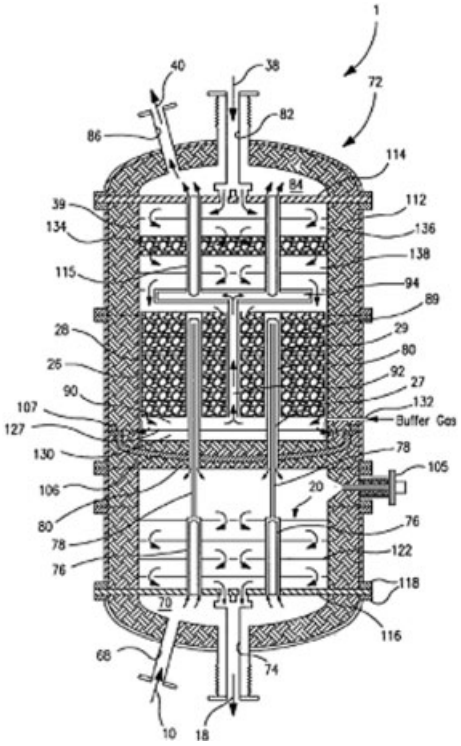


Figure 8.11 Figure 2 from U.S. Patent Application 20030039601 A1 [26]. Longitudinal cross-sectional view of a reactor (72) for tubular ceramic membranes (80).

each tube in a bayonet style arrangement. Unlike the Linde AG design shown in Fig. 8.9, the fresh air feed to the Praxair tubes is fed to the tubes through the bayonet instead of being withdrawn through the bayonet. Another difference between the Praxair design and the Linde AG design of Fig. 8.9 is that in the Praxair design, through the use of baffles, the syngas can be directed in cross flow across the tubes, such that the syngas flow can pass over multiple tubes in series flow. Several conceptions of these baffle arrangements are shown in Fig. 8.12. This innovation allows the incremental conversion of the syngas over each tube, addressing the challenge of the potentially large temperature gradient along a tube performing full conversion and the control issues involved in full conversion along a single tube. Therefore, the Praxair design attempts to solve the same challenges as the Linde AG design of Fig. 8.10 without resorting to multiple penetrations on the vessel diameter.

The Praxair design of Fig. 8.11 also incorporates integral heat exchange inside the pressure vessel, heat exchange sections being located at the top and bottom of the vessel. The fresh air feed is preheated against the hot oxygen-depleted air exhaust in the section of the reactor at the bottom of the figure. A second heat exchanger is located at the top of vessel and allows the methane feed to be preheated against the hot syngas product stream. The ceramic membranes and the heat exchangers are designed such that each of the components can freely expand

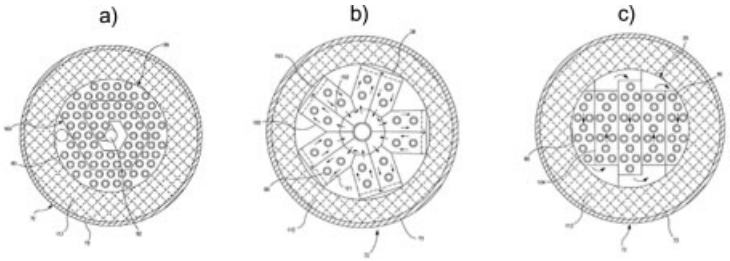


Figure 8.12 Figures 3 (A), 4 (B), and 5 (C) from U.S. Patent Application 20030039601 A1 [26]. Cross-section views of reactors (72) for tubular membranes showing possible arrangements of baffles (100) to produce series flow of the syngas across the membrane tubes (80).

or contract. This is a potentially important consideration since different materials are used in each of these components, and all of these components expand and contract significantly during heating and cooling of the reactor. Like the Linde design of Fig. 8.9, the Praxair vessel incorporates an internal tube sheet (labeled 106 in Fig. 8.11) that separates the high pressure syngas atmosphere from the low pressure air atmosphere. The ceramic membrane tubes pass through this tube sheet. An effort is made to reduce the metal temperature of this tube sheet by insulating this element from the hot syngas and hot air atmospheres on either side of it. A potential safety concern of high pressure syngas leaking into the low pressure air stream at the ceramic-to-metal seal (labeled 126) is mitigated by including a second ceramic-to-metal seal (labeled 128). A high pressure buffer gas, such as carbon dioxide, steam or nitrogen is fed into the space between the two seals. Since there is little or no pressure differential between the syngas product and the buffer gas, there will be little driving force for leaks of the syngas into the buffer gas. Any leakage of the high pressure buffer gas across seal 126 into the air stream, while undesirable, does not create any safety concerns. This arrangement reduces the need for the ceramic-to-metal seals to have very low leak rates.

8.3
Planar Membrane Reactors

The consortium led by Air Products and Chemicals, Inc. has developed designs for reactors using planar membranes. Unlike the case of the shell and tube heat exchanger inspiring tubular membrane reactor designs, there are fewer analogous heat exchange technologies on which designers of planar membrane reactors can build. A conceptual design from the Air Products consortium is shown in Fig. 8.13. This design uses a horizontal cylindrical pressure vessel with the planar membranes filling the interior [27]. A longitudinal view is shown in Fig. 8.14.

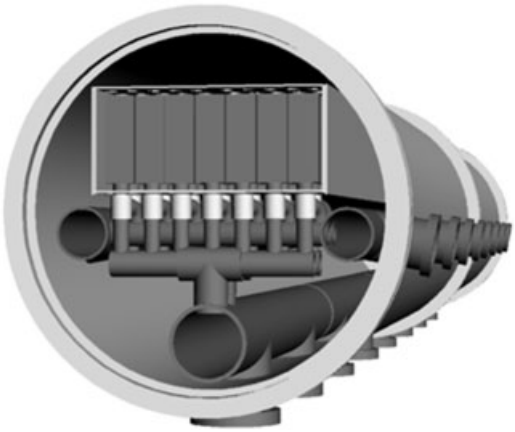


Figure 8.13 Cross-section of a conceptual syngas reactor for planar membranes showing the stacks of planar membranes (top), air feed manifolding to the stacks (middle) and the nonpermeate exit manifolding (bottom).

Stacks of planar membranes are arranged such that the syngas is in series flow over the stacks. Each stack consists of a number of planar membranes over which the syngas passes in parallel. In Fig. 8.14, each stack is manifolded to fresh air feed and spent air return pipes. The design in this figure has a fresh air feed to each stack, although other designs where the air makes passes through multiple stacks can also be imagined.

Since the syngas passes over multiple stacks arranged in series, it is only partially converted over each stack and temperature and composition control can be more easily achieved. This is similar to the design intent of the Linde reactor

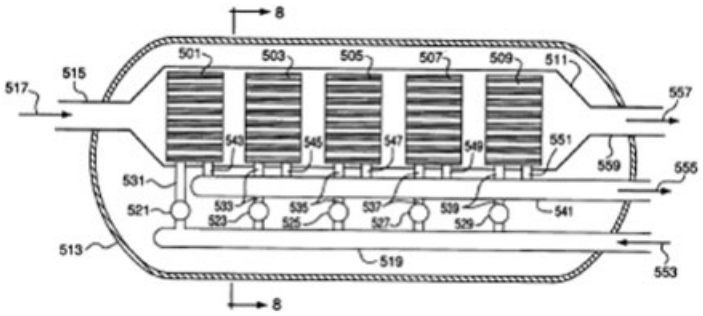


Figure 8.14 Figure 5 from U.S. Patent Application 20050031531 A1 [27] showing a longitudinal cross-section of a reactor for planar membranes, and showing the stacks of membranes (501, 503, etc), the reactant feed (517),the syngas product (557), the fresh air feed (553), the nonpermeate (555) exhaust (555).

using tubes perpendicular to the syngas flow. The temperature and composition gradient across any single membrane is also much smaller than if full conversion was achieved across a single membrane. By reducing the temperature and composition gradient across a single membrane, the potential mechanical stresses due to the thermal and composition gradients are also reduced. Unlike the Linde design of Fig. 8.10, the Air Products design, shown in Fig. 8.14, only has four vessel penetrations: nozzles for the fresh syngas feed and syngas product at each end of the reactor; and feed air and oxygen depleted air pipes. Increasing the size of the reactor does not increase the number of penetrations, unlike the Linde AG design in which the number of vessel penetrations increases with increasing reactor size. Manifolding of the air feed and spent air exhaust to each stack is done inside the vessel. This reduces the number of vessel penetrations to the minimum, potentially saving on vessel fabrication costs when compared to the design in Fig. 8.10.

As shown in Fig. 8.14, the membrane stacks are arranged in a flow duct inside the Air Products reactor. The flow duct contains the hot syngas and channels the syngas to flow through the membrane stack modules preventing the syngas from bypassing around the stacks. Insulation may be provided either on the inside surface of the vessel, the interior or exterior of the flow duct or some combination of the three options. By having internal insulation of the vessel, the reactor vessel itself remains relatively cool. By reducing the temperature of the metallurgy of the vessel, high temperature superalloys are not required, and less expensive metals can be used for the reactor. The four penetrations where the air is fed and withdrawn and the syngas is fed and the product is withdrawn can be internally insulated also, keeping the metal temperatures at each of these penetrations low. If insulation is used inside or outside the flow containment duct, the temperature of the air manifold piping shown in Fig. 8.14 inside the vessel will be reduced also. These air pipes can also be internally insulated, further reducing the metal temperature. Again, lowering the metal temperature decreases the need to use expensive high temperature alloys and allows the use of less expensive metallurgy.

One of the most attractive features about a planar membrane design is that it is generally possible to achieve a higher ratio of membrane area to unit volume of reactor than for tubular membranes. The planar membranes in a stack can be spaced close together, and the stacks themselves can be placed as close as necessary without the ceramic-to-metal seals becoming too close. This results in a high amount of membrane area in a cross section of reactor. Even though some amount of the reactor cross section is occupied by the air feed and nonpermeate manifold piping and the space between the square flow duct and the round vessel inside diameter is unused, the high packing density of the planar membranes can still potentially result in more membrane area per unit volume of reactor than a tubular design.

8.4
Ceramic-to-Ceramic Seals

One of the key enabling technologies required to implement planar or monolithic membranes are robust ceramic-to-ceramic seals. To enjoy all of the benefits afforded by the planar design, it is necessary to be able to stack the planar membranes into multimembrane stacks. Ceramic fabrication technology would typically be able to produce only single sintered planar membranes. Therefore, ceramic-to-ceramic seals are required to attach the sintered planar membranes to each other to form the stack. For monolithic designs, means of attaching the ceramic manifolds to the monolith are required. The requirements for the seals for the planar or monolithic membranes are quite significant. The seals must reliably separate high pressure syngas from low pressure air. The combustion due to any leaks in the seal could produce excessive localized heating of a membrane, which could result in excessive stress and failure of the ceramic. The material used for the seals must be stable in both the hot oxidizing environment on the air side of the membrane and the hot reducing environment on the syngas side of the membrane. The syngas stream may also contain up to 50–60% steam, which places its own demands on the material used to seal the ceramic parts together, since many metal hydroxides are more volatile than their corresponding metal oxides [28]. Not only must the seal material meet these technical requirements, the number of seals required in the stacks of planar membranes in a syngas reactor means that any solution must use economic materials and sealing processes.

Three different sealing technologies have been proposed for ceramic-to-ceramic seals: glass or glass-ceramic seals, braze seals, and all-ceramic seals. The first two types are well known and widely used in the ceramics industry, although the experience base is with different ceramics and typically different use conditions than the materials and conditions used for ceramic syngas membranes. With the decades of industrial experience with glasses, glass-ceramics and brazes, these options have received the most attention in the patent and academic literature. However, the all-ceramic seals developed by the Air Products consortium offer a number of important advantages over the glass or braze technologies.

Silica, boron, and phosphate-based glasses and glass-ceramics have been used for the seals in a variety of ceramic systems. One of the attractive features of glasses is that the glasses or glass-ceramics easily conform to the sealing surfaces when the glass softens at the sealing temperature. This reduces the requirement that the sealing surfaces be flat and parallel. Additionally, at operating temperature, a glass or a glass phase has the potential to relax any stresses at the seal due to viscous flow of the glass. This allows any stresses due to slight thermal expansion mismatches between the seal material and the ceramics to be joined to relax. Glasses also allow a wide range of stoichiometry variations and cation substitutions. Thus, it may be possible to tailor the coefficient of thermal expansion of a glass by modifying its composition so that it can match the ceramics to be joined. These features have allowed seal developers to devise several systems of glass and glass-ceramic families that have been suggested for application for ceramic syngas membranes.

Although glass and glass-ceramic seals possess the advantage of being well understood due to their long history of industrial use, some of their features that are counted as advantages in other applications become liabilities when the materials are used in seals for ceramic syngas membranes. In particular, the high operating temperature of the syngas membrane, coupled with the high pressure difference across the seal and the high steam partial pressure in the reducing syngas environment, make it difficult to employ many glass compositions for seals in a ceramic syngas membrane. The three most common glass formers in glasses are boron, silica and phosphorus. Each of these components is somewhat volatile at syngas membrane operating conditions. Phosphate-based glasses are unsuitable for continuous use at high temperatures due to the volatility of P_2O_5 and the tendency for the phosphates to crystallize to form compounds that have low stability under the reducing, high steam partial pressure conditions found on the permeate side of a ceramic syngas membrane [29]. Boron-containing glasses suffer from a similar problem. Under the high steam and high hydrogen partial pressures that are found on the syngas side of the membrane, $B_2(OH)_2$ and $B_2(OH)_3$ vapors will form, leading to loss of seal material and corrosion of the seal. Even silica will vaporize as $Si(OH)_4$ in high partial pressures of steam at elevated temperatures [30]. Therefore, from a thermodynamic perspective alone, typical glass compositions will be challenged to resist corrosion and remain leak free in service.

The mechanical properties that make glasses attractive in most other applications, namely the ability to flow to relax stresses and to conform to the sealing surfaces, also become significant disadvantages in ceramic syngas membrane service. The high pressure differential across the seal will exert a constant stress on the seal. The seal must be able to withstand the pressure gradient without deforming or leaking. However, at the high operating temperatures of the syngas membrane, glass phases will have some viscosity and will flow under the applied hydrostatic stress, eventually resulting in leakage of the high pressure syngas into the low pressure air. One solution proposed by researchers at Pacific Northwest Laboratories is to crystallize the glass into a ceramic material [31, 32]. These glass families contain alkaline earth, alumina, silica and boron, allowing thermal expansion coefficients up to 13 ppm $°C^{-1}$. This would allow expansion matching to some, but not all, of the ceramics that have been proposed for syngas membranes. The innovation of these researchers is that the phases formed after crystallizing these glasses have similarly high expansion values to those of the glass phases. This also addresses the potential problem with glasses eventually crystallizing into a ceramic at the syngas operating temperature. The invention disclosed in their patent [31] describes a composition that maintains an essentially constant coefficient of thermal expansion in both the glass and crystalline phases. This enables a seal to be made at one temperature, crystallization of the seal into a ceramic at a second temperature and operations at a third temperature, without placing the seal and ceramic membrane components under stress due to a mismatch in expansion properties during the thermal processing and operation. This solution also at least partially addresses the potential issue with volatility of the glass components. Silicon or boron in a crystalline phase is likely to have a lower

thermodynamic activity than in a glass phase. This will lower the vapor pressure of any volatile silicon or boron species. Although crystallization will not eliminate the volatility issue entirely, it may reduce the vapor pressure enough to give the glass-ceramic an improved lifetime.

One of the truly revolutionary technologies that have been developed to seal ceramic membrane components to each other is the sealing technologies developed by Butt et al. at Ceramatec and Air Products [33, 34]. Their inventions are methods to form an all-ceramic seal without the use of any glass phases. Their inventions use a seal material that will melt, or at least contains a phase that will melt, at a temperature below the sintering temperature of the ceramics to be joined. During thermal processing, the seal material forms a transient liquid phase when the seal temperature exceeds the melting temperature of the seal material. Once a liquid phase forms, the seal material can conform to the sealing surfaces. The liquid phase can react with any solid portion of the seal material and form a solid reaction product which becomes the joint. Interdiffusion with the ceramics to be joined can also occur. One of the foremost examples of this technology is the process of joining two sintered bodies (flat plates and/or tubes) of $(La_{0.75}Ca_{0.25})_{1.01}FeO_{3-x}$, having a sintering temperature of 1450 °C. The seal material was formed from a mixture of two compositions. The first composition was 20 wt% CaO–80 wt% Fe_2O_3 which has a melting point of approximately 1205 °C [35]. The second composition 84.2 wt% La_2O_3–15.8 wt% Fe_2O_3, which has a melting temperature in excess of the sintering temperature of the $(La_{0.75}Ca_{0.25})_{1.01}FeO_{3-x}$ ceramics that are to be joined. The seal material consisted of 2.07 parts by weight of the 84.2 wt% La_2O_3–15.8 wt% Fe_2O_3 composition to each part by weight of the 20 wt% CaO–80 wt% Fe_2O_3. The seal material was applied between two tubes of $(La_{0.75}Ca_{0.25})_{1.01}FeO_{3-x}$. The parts were heated to a temperature of 1250 °C, which is above the melting point of the CaO–Fe_2O_3 compound and below the sintering temperature of the bodies to be joined. The parts were held at this temperature for one hour, which was sufficiently long to allow formation of the liquid phase and reaction of the liquid phase with the solid phase and the surfaces of the parts to be joined. In this reaction, a third multicomponent metallic oxide, $(La_{0.75}Ca_{0.25})_{1.01}FeO_{3-x}$, is formed at the seal. Since the third multicomponent metallic oxide is identical to the composition of the ceramics to be joined, the thermal expansion coefficient of the joint material closely matches the thermal expansion coefficient of the $(La_{0.75}Ca_{0.25})_{1.01}FeO_{3-x}$ parts that were joined. The joint thus has about the same thermal cycling stability as the parts themselves. The joint also has the same chemical properties as the membrane material and therefore, the seal will be as resistant to the process atmospheres as the membrane material itself. There are also no concerns with reactivity or chemical incompatibility of the seal material and the membrane material since the membrane and the joint material are chemically identical after the joint has been made. The invention of Butt et al. [33, 34] thus produces a ceramic-to-ceramic seal that has the very attractive properties of being expansion matched to the membrane material; chemically identical to the membrane material; just as chemically compatible with the process gas atmospheres as the membrane material; just as thermodynamically

stable as the membrane material; made from materials costing no more than the membrane material; and straightforward to fabricate. Their invention is one of the key enabling technologies for implementing planar ceramic membranes.

A third type of seal that has been used for syngas membranes, primarily in academic settings, is the braze seal. Like the other types of seals, braze seals are exposed to the reducing, high steam partial pressure syngas atmosphere and the oxidizing air atmosphere. The only commonly used braze metals that are stable in the hot oxidizing atmosphere are the precious metals Au, Ag and Pd. Silver has a melting temperature of 963 °C. Silver could only be used as a component in a higher melting alloy for at least some syngas applications since the patent literature quotes membrane operating temperatures of $\geq$1000 °C [36]. In addition, loss of silver due to volatilisation at the higher temperatures would become an important concern. The expense of seals containing gold or palladium would have a negative impact on the overall economics of a commercial reactor unless care was taken to minimize the amount of sealing area required. For these reasons, there has been little documentation of the use of braze seals on larger scale syngas reactors.

8.5 Ceramic-to-Metal Seals

One of the significant challenges facing a developer of ceramic syngas membranes is the need to connect the ceramic membranes to the air feed and air exhaust manifolds or tube sheets. These seals are needed not only for the eventual commercial deployment of syngas reactors with dense ceramic membranes but also to progress the technology from the laboratory scale to the pilot scale and precommercial stages. As a result, ceramic-to-metal sealing technology for ceramic membranes has received considerable attention in the patent literature. The demands on the seal are considerable. The seal will typically operate at elevated temperature and possibly at the membrane operating temperature. These seals must also withstand the pressure differential between the low pressure air and the high pressure syngas and prevent leakage of the syngas into the air. Finally, either the ceramic and metal components must be expansion matched, or the seal must be tolerant of at least some expansion mismatch between the ceramic and metal components. There are no commercially available ceramic-to-metal seals that meet these criteria, inspiring a number of individuals and organizations to develop their own novel designs. The more successful of these designs may eventually find application in fields other than dense membranes.

The ceramic-to-metal seals for dense ceramic syngas membranes that have been suggested or demonstrated fall into one of three categories:

- Glass based seals
- Braze seals
- Mechanical seals

The glass-based and braze seals have the potential to form hermetic seals, but it can be difficult to expansion match the ceramic, metal, and seal compositions. Glasses also potentially suffer from some of the same issues that made ceramic-to-ceramic sealing with glasses challenging, namely the volatility of the glass formers in high partial pressures of steam and the crystallization of the glass during operation. Brazes also face the same technical and economic issues that affect their implementation as ceramic-to-ceramic seals, such as the limited material choices for use in a hot oxidizing environment and the high cost of the precious metals used in brazes. Nonetheless, there are ample examples of ceramic-to-metal seals using glass formers and brazes that have been successfully demonstrated. Whether these two technologies advance sufficiently to find commercial application remains to be seen.

Glass-based seals come in a variety of forms. At least two organizations have developed ceramic-to-metal seals based on a mixture of a glass former with the membrane material itself. Workers at Argonne National Laboratories have demonstrated a glass-based ceramic-to-metal seal where a tube of $SrFeCo_{0.5}O_x$ was joined to Inconel 600 alloy [37]. The seal consisted of a mixture of the glass former B_2O_3, SrO and the membrane material itself, $SrFeCo_{0.5}O_x$. After forming the seal at 1155 °C and cooling to room temperature, the seal was demonstrated to hold an external pressure of 200 psig (1.4 MPa). At 900 °C, the seal could hold an external pressure of 5 psig (34 000 Pa) for 48 h. A similar approach was taken by researchers at Eltron Research Inc. Their patented ceramic-to-metal seal consists of a mixture of the glass former silica, along with the ceramic membrane composition and three or more metal oxides [38]. No performance data were given for this seal.

There are a number of mechanical ceramic-to-metal seal concepts that have been published that use a gasket, gland, packing or other compliant member to seal between the ceramic and metal component. Although these types of seals have been successfully used in lower temperature applications for decades, the unique aspects of these new concepts is that the seals are being used at high temperature to seal between dissimilar materials. Examples of seals using a gland are the seals developed by Praxair [39] and shown in Fig. 8.15. A sealing material is placed between the conical foot of the ceramic tube "2" and the metal member "22." The sealing material may be a rope such as NEXTEL™ ceramic fiber rope, a conical gasket such as a ceramic fiber mat or felt, or a graphite sheet. Tightening the metal component compresses the sealing material, which in turn holds the ceramic element in place. A second sealing material at the base of the tube may be used in addition to the sealing material placed on the diameter of the tube. Labeled "38" in Fig. 8.15, this gasket can be made from a graphite core jacketed with a metal jacket such as Inconel. The patent also suggests that a washer or coating be placed between the ceramic tube and the gasket "38" to avoid chemical interaction between the metal jacket of the gasket with the ceramic. Possible choices suggested for the washer include refractory ceramic washers such as magnesia, ceria or zirconia, or metal washers made from gold, silver or nickel–chromium alloy.

A second type of ceramic-to-metal seal patented by Air Products uses compliant, metallic, toroidial rings as the seal [40]. Figure 8.16 shows a ring in cross-section

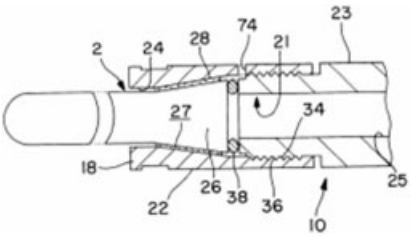


Figure 8.15 Figure 3 from U.S. Patent 6 547 286 B1 [39] showing a sealing material (38) placed between the conical foot of the ceramic tube (2) and the metal member (22). The sealing material may be a rope such as NEXTEL™ ceramic fiber rope, a conical gasket such as a ceramic fiber mat or felt, or a graphite sheet.

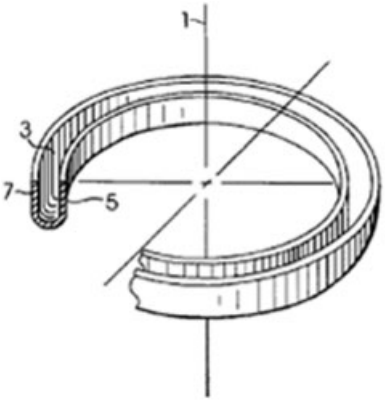


Figure 8.16 Cross-section of a toroidial ring from Fig. 1 of U.S. Patent 6 302 402 [40].

and Fig. 8.17 shows a cross-section of a seal incorporating such a ring. The ring seals on the outside diameter of the ceramic tube 21 and the inside diameter of the metal member 19 in Fig. 8.17. Preferably, the ring has an interference fit with the ceramic and metal components, meaning the ring is in slight radial compression. The ring is compliant by virtue of its thin wall thickness and the choice of the metal. Typically, the high pressure gas, such as syngas, would be placed on the open end of the U-shaped seal and the low pressure gas such as air would be placed on the inside. At operating conditions, the pressure of the high pressure gas will help activate the seal by pushing the arms of the U-ring against the ceramic and metallic components of the seal, thereby improving the sealing action of the U-ring. The U-ring can be coated with a soft metal to improve the sealing efficiency. As the seal assembly is heated, differences in the thermal expansion of the metal and ceramic components of the seals can be accommodated by the compliant nature of the U-ring. When the seal assembly is cooled, the U-ring will spring back to maintain the seal.

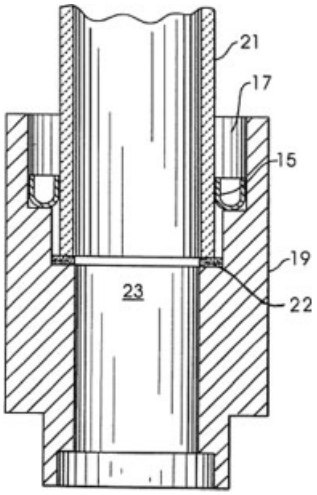


Figure 8.17 Cross-section of a ceramic to metal seal from Fig. 3 of U.S. Patent 6 302 402 [40], showing the torodial ring (15) sealing between a ceramic tube (21) and a metal housing (19).

Possible materials of construction for the U-ring include the nickel superalloys Inconel 600, Inconel 625, Incoloy 800 and Haynes 230. The soft coating that can be used on the U-ring can be gold or silver, or alloys containing gold or silver. In the examples provided in the patent, these seals showed very low leak rates, even when used against pressure differentials of up to 200 psig (1.4 MPa) at temperatures up to 850 °C. Therefore, seals of this type would have promise in the ceramic syngas membrane applications.

8.6 Summary and Conclusions

The potential of ceramic syngas membranes has been proven numerous times by laboratories all over the world. Economic analyses indicate the potential of a significant cost advantage for the ITM Syngas process over conventional syngas production processes [21]. To reap these benefits, developers of ceramic syngas membrane processes must make the transition from laboratory phenomena to a robust and proven chemical process. Just as the development of the membrane materials required major innovations, the road to scale-up and commercialization of ceramic syngas membranes will be paved with innovations. Membrane designs that meet the performance, economic and lifetime criteria are needed. Reactor designs that accommodate these membranes but still maintain the process's economic benefits need to be developed. Finally, ceramic-to-ceramic seals and ceramic-to-metal seals need to be invented. Three major industrial consortia have been formed to address these technical challenges. Much progress has been made in all of these areas, and the patent literature is continuing to document the extent of that progress. Although challenges still lie ahead for the developers of this technology, they have conquered many other problems and have momentum as they move ahead.

Acknowledgements

The author wishes to acknowledge and thank Joseph Abrardo, Douglas Bennett, John Bernhart, Christopher Chen, Ted Foster, Robert Iampietro, Michael Kempf, Alexander Makitka, Brenda Messersmith, Rick Meyers, Christopher Miller, Eric Minford, Leslie Schaffer, William Waldron, Matthew Watson, and Robert Weimer of Air Products and Chemicals, Inc.; Darryl Butt, Raymond Cutler, Charles Lewinsohn, James Steppan, Dale Taylor, and Merrill Wilson of Ceramatec; Ted Ohrn and Keith Rackers of SOFCo EFS; Michael Mundschau and Tony Sammells of Eltron Research Inc.; and Steve DeCanio, Karl Gerdes, Charles Kibby, Randy Roberts, and William Schinski of Chevron Energy Technology Co.; and David Shelleman and Richard Tressler of The Pennsylvania State University for their help and assistance.

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